

Evaluating MAKSA Mobile App's Accuracy in Assessing Motility of Fresh Bovine Semen: A Comparative Study with Computer-Assisted Sperm Analysis (CASA) (Okara, Pakistan 2026)

Date: February 2026

Location: Okara, Pakistan

Conducted by: Dr. Misbah Aslam (Okara, Pakistan)	Verified by: Avare Biotech Team (Dubai, UAE)
Lab technologist at Sexed sorted Semen Lab (Pakistan) I'm a vet specialized in animal reproduction.	Maksim Mershiev, The CEO&Co-Founder of Avare Biotech
	



Signature:

Signature:

Executive Summary / Conclusions

Smart phone based MAKSA technology demonstrated an ease in Fesh Bovine semen Evaluation.

Objective

To validate the smart phone based MAKSA (Mobile-Assisted Key Semen Analysis) technology for **Fresh Bovine Semen Analysis**, showing good agreement with CASA (Androvision miniTube) for total motility, progressive motility, and concentration (NucleoCounter), indicating MAKSA's potential as a reliable on-farm tool.

Materials and Methods

Table 1. MAKSA results comparison					
Tests done	Key Semen Parameters	MAKSA	CASA Motility	NucleoCounter Conc.	Difference %
Bull 1 (Auburn Rojo)	Concentration (M/ml)	1740.7		1356	-22
	Total Motility (%)	84	71.22		-15.21
	Progressive Motility (%)	72	70.39		-2.23
Bull 2 (Braford)	Concentration (M/ml)	921.2		1002	8.77
	Total Motility (%)	79	82.16		4
	Progressive Motility (%)	75	82		9.33
Bull 3 (Gabru)	Concentration (M/ml)	464.1		389.5	-16.07
	Total Motility (%)	82	68.67		-16.25
	Progressive Motility (%)	73	68.63		-5.98
Bull 4 (SK-1 Gulabi)	Concentration (M/ml)	1162.2		912	-21.52
	Total Motility (%)	80	77.38		-3.27
	Progressive Motility (%)	76	76.4		0.52





Bull 5 (Janos Manso)	Concentration (M/ml)	1420.7		921	-35.17
	Total Motility (%)	81	69.74		-13.90
	Progressive Motility (%)	71	69.74		-1.76
Bull 6 (Lotto 062M)	Concentration (M/ml)	464.1		405	-12.73
	Total Motility (%)	82	72.03		-12.15
	Progressive Motility (%)	73	71.91		-1.49

Discussion / Expert Insights

- Total and progressive motility showed strong correlation between MAKSA and CASA readings.
- Concentration values differed between MAKSA and CASA, suggesting need for app calibration with fresh-diluted semen data.
- **MAKSA** is a valuable on-farm tool for quick semen analysis, with potential for improved accuracy with calibration enhancements.

Recommendations

- Add in-app calibration videos for fresh semen analysis.
- Include training mode with feedback for new users.
- Share field condition tips in the app for on-farm use.

Appendix

- Screenshot of the semen videos made and CASA results
- MAKSA PDF report samples





Bull 1: CASA readings

Avare Biotech



@Avareit



Sample Report

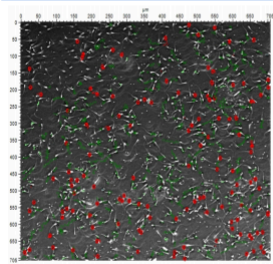
Donor name: AUBURN ROJO
Donor ID: 706
Life number: 706
Species: Bovine
Breed: Red Brahman

Minitüb GmbH
Hauptstr. 41
84154 Tistlitzbach
Germany

Internet:

Motility & Concentration of native ejaculate

Analysis conditions



Sample analysis

Date of analysis	2/26/26
Total no. of sperm analysed	2,184.00
Number of fields	3.00
Concentration	[10 ⁶ /ml] 87.50
Total motility	[%] 71.22
Progressive motility	[%] 70.39
Progressive fast motility	[%] 58.95
Progressive slow motility	[%] 11.44
Progressive circular motility	[%] 0.00
Local circular motility	[%] --
Local motility	[%] 0.83
Immotile	[%] 28.78

	VCL [µm/s]	VSL [µm/s]	VAP [µm/s]	DCL [µm]	DSL [µm]	DAP [µm]	ALH [µm]	BCF [Hz]	HAC [rad]	WOB (VAP/VCL) (VSL/VCL) (VSL/VAP)	LIN	STR
Average native	167.69	70.56	82.10	28.64	10.62	13.12	1.57	11.30	0.37	0.49	0.42	0.86

Bull 1: MAKSA values



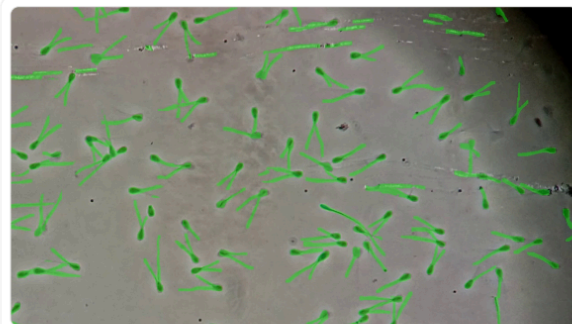
@avare_biotech



@AvareBiotech

Analysis 11.03.2026

Animal: — Name: Auburn Rojo 706 Age: 3 Breed: Red Brahman Status: Ready Created: 2026-03-11 01:17



Total objects:	235
Concentration (millions/ml ^ 3):	1740.70
Dilution rate:	[1:11]
Volume of field of view (mm^3):	
Progressive Motility:	72%
Non-Progressive Motility:	12%
Immotility:	16%

Agglutination

Bull 2: CASA readings

Bull 2: MAKSA values



<http://avareit.com>



UAE, Dubai, DIP1, The European
Business Centre, Office #170



+971 50 641 2775

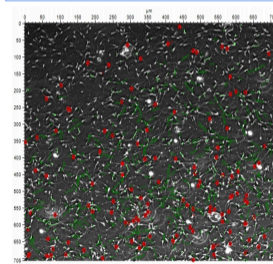


Sample Report

Donor name	braford	Species	Bovine
Donor ID	3567 dayee	Breed	Braford
Life number			

Motility & Concentration of native ejaculate

Analysis conditions



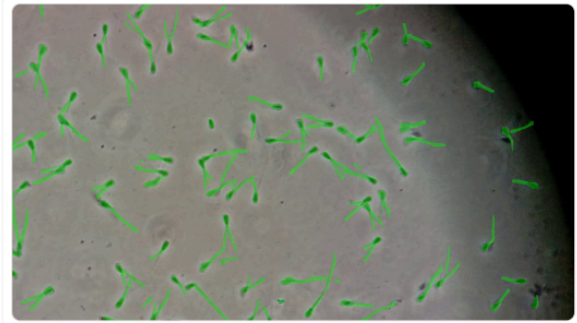
Sample analysis

Date of analysis	2/25/26
Total no. of sperm analysed	1,971.00
Number of fields	3.00
Concentration	[10 ⁶ /ml] 78.96
Total motility	[%] 82.16
Progressive motility	[%] 82.00
Progressive fast motility	[%] 58.54
Progressive slow motility	[%] 23.46
Progressive circular motility	[%] 0.00
Local circular motility	[%] --
Local motility	[%] 0.16
Immotile	[%] 17.84

	VCL [µm/s]	VSL [µm/s]	VAP [µm/s]	DCL [µm]	DSL [µm]	DAP [µm]	ALH [µm]	BCF [Hz]	HAC [rad]	WOB (VAP/VCL)	LIN (VSL/VCL)	STR (VSL/VAP)
Average native	131.65	57.59	66.24	30.74	12.46	14.91	1.25	15.85	0.30	0.50	0.44	0.87

Analysis 11.03.2026

Animal: — Name: Braford 3567 Age: 3 Breed: Braford Status: Ready Created: 2026-03-11 02:35



Total objects:	148
Concentration (millions/ml ³):	921.20
Dilution rate:	[1:11]
Volume of field of view (mm ³):	
Progressive Motility:	75%
Non-Progressive Motility:	4%
Immotility:	20%

Agglutination

Found:	No
--------	----

Bull 3: CASA readings

Bull 3: MAKSA values





Minitüb GmbH

Hauptstr. 41
84184 Tiefenbach
Germany

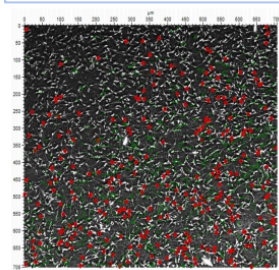
Internet:

Sample Report

Donor name	GABRU	Species	Bovine
Donor ID	001	Breed	Cholistani
Life number			

Motility & Concentration of native ejaculate

Analysis conditions



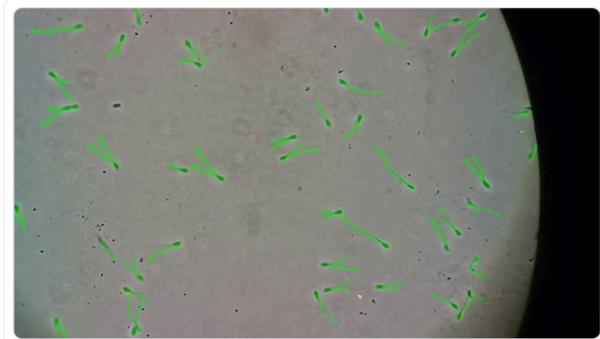
Sample analysis

Date of analysis	2/24/26
Total no. of sperm analysed	2,953.00
Number of fields	3.00
Concentration	[10 ⁶ /ml]
	118.30
Total motility	[%]
	68.67
Progressive motility	[%]
	68.63
Progressive fast motility	[%]
	51.96
Progressive slow motility	[%]
	16.67
Progressive circular motility	[%]
	0.00
Local circular motility	[%]
	--
Local motility	[%]
	0.04
Immotile	[%]
	31.33

	VCL	VSL	VAP	DCL	DSL	DAP	ALH	BCF	HAC	WOB	LIN	STR
	[µm/s]	[µm/s]	[µm/s]	[µm]	[µm]	[µm]	[µm]	[Hz]	[rad]	(VAP/VCL)	(VSL/VCL)	(VSL/VAP)
Average native	143.63	61.15	70.96	21.92	8.26	10.21	1.39	10.80	0.34	0.49	0.43	0.86

Analysis 11.03.2026

Animal: — Name: Gabru 001 Age: 3 Breed: Cholistani Status: Ready Created: 2026-03-11 02:43



Total objects:	88
Concentration (millions/ml ³):	464.10
Dilution rate:	[1:11]
Volume of field of view (mm ³):	
Progressive Motility:	73%
Non-Progressive Motility:	9%
Immotility:	17%

Agglutination

Found: No

Bull 4: CASA readings

Bull 4: MAKSA values



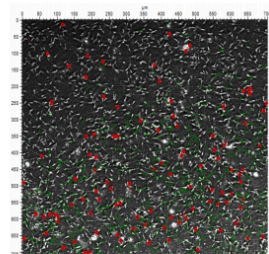


Sample Report

Donor name	SK1- GULABI CHOLISTANI	Species	Bovine
Donor ID	009	Breed	Cholistani X
Life number			

Motility & Concentration of native ejaculate

Analysis conditions



Sample analysis

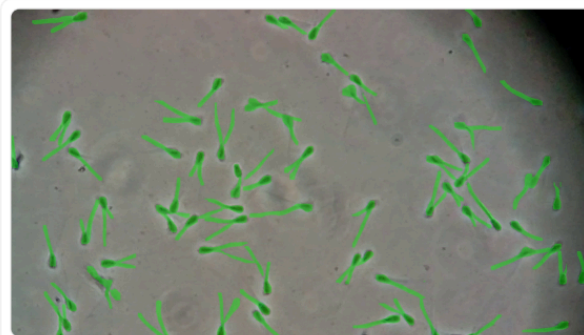
Date of analysis	2/25/26
Total no. of sperm analysed	1,879.00
Number of fields	3.00
Concentration	[10 ⁶ /ml] 75.28
Total motility	[%] 77.38
Progressive motility	[%] 76.40
Progressive fast motility	[%] 50.14
Progressive slow motility	[%] 26.26
Progressive circular motility	[%] 0.00
Local circular motility	[%] --
Local motility	[%] 0.98
Immotile	[%] 22.62

	VCL	VSL	VAP	DCL	DSL	DAP	ALH	BCF	HAC	WOB	LIN	STR
	[µm/s]	[µm/s]	[µm/s]	[µm]	[µm]	[µm]	[µm]	[Hz]	[rad]	(VAP/VCL)	(VSL/VCL)	(VSL/VAP)
Average native	117.15	54.07	61.49	23.23	9.80	11.72	1.03	15.66	0.26	0.52	0.46	0.88

Analysis 11.03.2026

Animal: — Name: SK1 Gulabi Cholistani Age: 3 Breed: Cholistani Cross Status: Ready

Created: 2026-03-10 23:02



Total objects:	119
Concentration (millions/ml ³):	1162.20
Dilution rate:	[1:11]
Volume of field of view (mm ³):	
Progressive Motility:	76%
Non-Progressive Motility:	4%
Immotility:	20%

Agglutination

Found:	No
--------	----

Bull 5 : CASA readings

Bull 5: MAKSA values



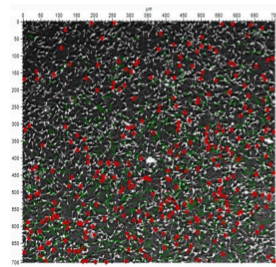


Sample Report

Donor name	JANOS MANSO	Species	Bovine
Donor ID	709	Breed	Grey Brahman
Life number			

Motility & Concentration of native ejaculate

Analysis conditions



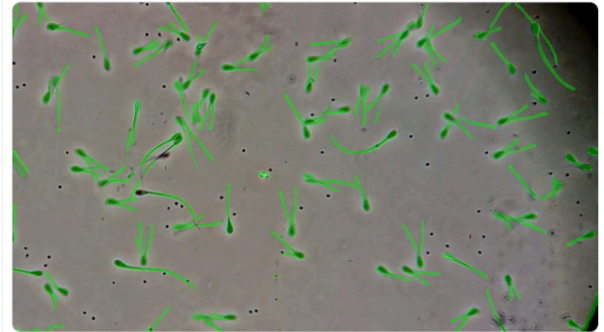
Sample analysis

Date of analysis	2/24/26
Total no. of sperm analysed	3,386.00
Number of fields	3.00
Concentration	[10 ⁶ /ml] 135.65
Total motility	[%] 69.74
Progressive motility	[%] 69.74
Progressive fast motility	[%] 62.50
Progressive slow motility	[%] 7.24
Progressive circular motility	[%] 0.00
Local circular motility	[%] --
Local motility	[%] 0.00
Immotile	[%] 30.26

	VCL	VSL	VAP	DCL	DSL	DAP	ALH	BCF	HAC	WOB	LIN	STR
	[µm/s]	[µm/s]	[µm/s]	[µm]	[µm]	[µm]	[µm]	[Hz]	[rad]	(VAP/VCL)	(VSL/VCL)	(VSL/VAP)
Average native	189.03	77.89	89.59	25.83	9.43	11.57	1.62	12.62	0.43	0.47	0.41	0.87

Analysis 11.03.2026

Animal: — Name: JANOS MANSO 709 Age: 3 Breed: Grey Brahman Status: Ready Created: 2026-03-11 02:39



Total objects:	175
Concentration (millions/ml ^3):	1420.70
Dilution rate:	[1:11]
Volume of field of view (mm ^3):	
Progressive Motility:	71%
Non-Progressive Motility:	10%
Immotility:	20%

Agglutination

Bull 6: CASA readings

Bull 6: MAKSA values



